

Power generation expansion pathways: A policy analysis of the Cameroon power system

Yvan Ayuketah^{a,b,*}, Samuel Gyamfi^{a,b}, Felix Amankwah Diawuo^b, Athanasios S. Dagoumas^c

^a Regional Center for Energy and Environmental Sustainability (RCEES), P. O. Box 214, Sunyani, Ghana

^b School of Energy, University of Energy and Natural Resources (UENR), P. O. Box 214, Sunyani, Ghana

^c School of Economics, Business & International Studies, University of Piraeus, 80 Karaoli & Dimitriou Str., 18534, Piraeus, Greece

ARTICLE INFO

Keywords:

Electricity generation
Emission targets
Power losses
Demand-side energy efficiency
Cameroon

ABSTRACT

The Cameroon electricity sector is currently plagued by inadequate generation capacity. The electricity supply deficit has had dire consequences on attaining the country's industrialization, socioeconomic and developmental objectives enshrined in her Vision 2035. However, there are sustainable solutions that minimize the required new capacity planned by the government. This paper analyses the implications of these sustainable policies on the electricity generation system of Cameroon. An adaptation of the Schwartz's methodology and the Low Emissions Analysis Platform (LEAP), integrated with the Next Energy Modelling system for Optimization (NEMO) was used to develop least-costs scenarios of Cameroon from 2016 to 2045. These scenarios are based on the key factors that would influence the electricity supply/demand of Cameroon. In addition to the business-as-usual scenario which defines the least-cost pathway with no policy interventions, the three alternative scenario categories are: power losses, demand side energy efficiency measures and carbon dioxide emission targets. Results show the alternative scenarios have significant benefits in avoided installed capacity, financial savings and avoided greenhouse gas emissions. This is emphasized by capacity savings of up to 44%, translating to possible economic benefits of up to 25% depending on government's commitment to reducing power losses. Furthermore, potential for emission trading is limited, although it becomes more appealing under a zero-carbon emission transition target in the electricity system. Therefore, the state should prioritize the reduction of power losses and enacting energy efficiency policies if it aims to safely and reliably minimize the energy needed for its economic growth.

1. Introduction

Cameroon in 2009 outlined her long-term vision to become an emerging economy by 2035 [1]. This plan called for a radical growth in her economy over the next 25 years. This growth should be sustained with marked socioeconomic improvement, structural reforms, and a revolution in the industry and agriculture sectors [1]. Actions towards attaining the vision objective has seen a spike in electricity demand. According to projections by the sector regulator, the Electricity Sector regulatory Agency (ARSEL), total electricity demand of Cameroon grows at an average of 6.7% annually [2]. In contrast, generation capacity from 2009 has increased at a comparatively slower rate, average of 3.7% annually between 2009 and 2020 [3]. Cameroon intended achieving 3000 MW by 2020 [2,3]. However, the latest report by the Ministry of Economy, Planning and Regional Development (MINEPAT) shows a

national installed capacity of 1650 MW in 2020 [3]. This represents a shortfall of 1350 MW compared to the envisioned 2020 target. Over the years, the generation-demand imbalance, has been the primary cause of electricity crisis in the nation.

The government has prioritized the development of hydropower and natural gas plants as the main solution to solving the electricity demand-supply deficit [1]. This is reiterated in earlier power sector masterplans like the Growth and Employment Strategy Paper [4], the Electricity Sector Development Plan 2035 (PSDP) and the Energy Sector Development Project (PDSN) which focused on developing large-hydropower and maximization of the nation's abundant natural gas potential [2]. Aspirations to provide electricity to rural and remote villages saw the formulation of the Rural Electrification Master Plan (REMP) [5]. The REMP targets the electrification of over 660 villages through a combination of off-grid systems (diesel, solar PV, mini-hydro, biomass and

* Corresponding author. Regional Center for Energy and Environmental Sustainability (RCEES), P. O. Box 214, Sunyani, Ghana.

E-mail address: yvanayuketah@gmail.com (Y. Ayuketah).

<https://doi.org/10.1016/j.esr.2022.101004>

Received 31 May 2022; Received in revised form 6 October 2022; Accepted 13 November 2022

Available online 30 November 2022

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hybrids) and grid extension.

Cameroon has historically been an insignificant greenhouse gas (GHG) emitter, contributing less than 0.02% to global carbon dioxide (CO₂) emissions [6]. However, the continual consumption of traditional biomass such as firewood and charcoal, (which currently make up over 70% of Cameroon's total energy consumption) for cooking will significantly increase the nation's GHG emissions [7]. The nation's development targets are justifiably expected to increase GHG emissions, mainly from the transport, power and residential sectors. Cameroon, as part of the Paris Agreement aim to reduce her total emissions by 32% compared to the business-as-usual scenario [8]. The power sector contributed to over 30% of Cameroon's 6.5 metric tons of carbon dioxide equivalent (MTCO_{2e}) total emissions in 2019. However, the planned redevelopment of thermal generation to bridge the supply gap, increase reserve margin and ensure generation adequacy will increase power sector emissions, contrary to its Nationally Determine Contributions (NDCs) [7].

From Cameroon's NDCs, specific emissions reduction measures for the power sector include the increase in energy efficiency and ensuring a 25% share of renewable energy in the total generation mix by 2035. Apart from the planned implementation of energy efficiency measures, the state's approach to meeting the electricity deficit has focused mainly on increasing generation capacity. Reducing electricity demand instead of increasing new capacity however provides a more sustainable alternative to both overcoming the electricity deficit, and reducing associated emissions. One strategy to reducing electricity demand entails the implementation of demand-side management policies, a requisite given Cameroon has no such policies in place. Another alternative is minimizing the high transmission and distribution losses incurred in the network. Furthermore, implementation of emissions taxes and emission trading mechanisms provide other methods for sustainably reducing the electricity demanded [9].

The need to maintain the stability of the grid, interconnection and coupling of the power system necessitates an integrated approach to evaluating sustainable pathways for the low-carbon energy transition. The peculiarities of each country (in terms of resource availability, policy objectives, economic strength etc.) have seen varying approaches to developing power generation pathways that ensure adequate supply while transitioning to a low carbon and efficient power system. Nonetheless, a common aspect is the use of scenarios [10]. Scenarios are plausible and alternative descriptions of the future [11]. Scenario analysis which seeks to assess the implications of various policies and assumptions on future energy system have been applied in literature. Examples of applied scenarios analysis include: the Brundtland Report [12], The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report [13] and mapping the "most technically feasible, cost-effective and socially acceptable" roadmap towards a net zero by 2050 for the global energy sector by the International Energy Agency (IEA) [14]. Another IEA study assessed the impact of the Covid-19 pandemic on the world energy outlook and clean energy transition prospects [15,16] assessed the optimal power generation expansion plan for Ghana, realizing renewables and energy efficiency could contribute up to 70% of required power generation. The future energy demand, supply and associated GHG emissions for Nigeria was explored by Ref. [17] using four scenarios. The results showed the energy demand and GHG emissions of Nigeria to reach 3075 PJ and 201.2 Mt Co_{2e} respectively by 2040. Furthermore, the industry and commercial sectors hold the most potential for energy efficiency, while the residential and transport sectors have the most potential for GHG emissions reduction. Another study by Ref. [18] assessed the impact of technological change on the energy system of Colombia, concluding energy efficiency efforts should be concentrated on the most relevant types of uses in each sector.

Studies applying scenario analysis to the Cameroon power system has been few and far between. Evaluation of the energy demand, optimization of the generation mix and quantification of associated emissions was performed by Ref. [19]. Another study by ARSEL projected the

electricity demand between 2014 and 2035 [2]. However, we observe several research gaps. Firstly, these studies were all done prior to 2016, so do not precisely represent the current reality of the Cameroon power system, nor account for power generation expansion trends. Secondly, the impact of environmental constraints such as carbon taxes on the generation system which could be necessary for the transition to a modern energy system have not been assessed. Last but not least, no study to be the best of authors' knowledge has assessed the economically efficient generation expansion pathway that ensures the generation-load balance and balanced reserve margin (15%–20% recommended [20]) for Cameroon.

This study seeks to fill the identified knowledge gaps while illuminating the least-cost development pathways available to Cameroon. The study also wishes to highlight the importance of improving the power network infrastructure and implementing energy efficiency measures. The novel aspect is that the study provides insights on four fundamental questions currently facing the Cameroon electricity sector and low-carbon energy transition:

- How would the generation mix evolve in the long-term, its associated costs and GHG emissions under stated policies and least-cost optimization conditions?
- Of the planned electricity sector policies/measures, what categories are more cost-beneficial in the long-run, and to what extent?
- How viable is carbon taxing and emission trading in Cameroon?
- What could be the potential future generation mix if a zero-carbon power system is targeted.

This study aims to trigger discussions between the state and stakeholders on the implications of policy tradeoffs and role of carbon pricing and emission trading on renewable energy adoption in the power system of Cameroon.

The rest of this study is divided as follows: Section II presents an overview of the electricity demand, generation and energy resources of Cameroon. Section III examine the scenario development methodology and modelling framework in addition to key assumptions and definition of the scenarios. In Section IV, the study results are presented and analyzed. Section V answers the question raised in Section I and discusses the strengths, novelty and limitations of the study. The policy implications and recommendations of the study are presented in the last section.

2. Cameroon power sector overview

The Republic of Cameroon operates three independent grids: the Eastern Grid (EG), the Northern Interconnected Grid (NIG) and the Southern Interconnected Grid (SIG). The Southern Interconnected Grid is the largest of the three grids, serving six of the ten regions of the country [21]. There is presently no interconnection between the grids, though a transmission line is currently being built to interconnect the SIG and EG. Plans also exist to interconnect the three grids, forming a unified grid by 2035. The transmission network, officially operated by the National Electricity Transmission Company (SONATREL) consists of 24 stations, connected by over 19445 km of high voltage, 15082 km of medium voltage and 15210 km of low voltage transmission lines. The distribution network, consisting of over 22608 km distribution lines is operated by the utility company Energy of Cameroon (ENEO) [22]. The nation of 26.55 million people has an electricity access rate of 70% in 2020. Rural and urban electrification rates were 32% and 98% respectively [23,24].

The total dependent installed capacity of Cameroon as of July 2019 is 1541.39 MW [21,25]. This composed of: hydropower (61%), natural gas (20%), oil (18%) and renewable energy sources (1%). The state-run utility ENEO owns 65%, which are made up of thirteen grid-connected and twenty-four remote thermal plants. The remaining 35% is owned by independent power producers (IPPs) [21]. Renewable energy

generation are mainly off-grid solar PV and small hydropower, the latter defined officially as less than 10 MW in Cameroon. Installed off-grid solar PV and small hydropower capacities respectively, were 14.19 MW and 0.3 MW [25]. Renewable energy sources are classified in Cameroon as: solar PV, wind, biomass and small hydropower.

Total electricity generation in 2019 was 7043 GWh. This constituted 74.7% from hydropower, 18.9% from natural gas 6.0% from oil and 0.4% from renewables. The historical energy generation of Cameroon is presented in Fig. 1 [21,25–27]. The share of grid-connected renewables in the generation mix can be considered insignificant, less than 0.5%. Grid-connected electricity consumption at the point of distribution in 2019 was 6544 GWh according ENEO [21]. This consumption was by the industry (56%), residential (21%), commercial (8%) and agricultural (1%) and other sectors (14%) [28]. The industry sector has historically been the highest consumer.

However, electricity supply in Cameroon is characterized by frequent (and increasing) brownouts and blackouts [19,21]. These outages often last more than 24 h s even in urban centers. The causes of the increscent network interrupts could be attributed to the high power losses. The network losses in turn stem from the dilapidated state of the three grids, their independency and little decentralization of generation plants, characteristics of a traditional grid. Another issue of the Cameroon power system is the absence of energy efficiency or demand side measures, which guarantees a safe, reliable and affordable option to reducing emissions and demand according to International Renewable Energy Agency (IRENA) [9]. There is also a reliance on hydropower which often causes annual electricity crisis due to drop in reservoir water levels in the dry season [29].

Like all countries in sub-Saharan Africa, the population of Cameroon has resorted to backup generators, commonly known as gensets, as a stopgap to the deficiencies of the grid. A study by the International Finance Corporation (IFC) [30] estimated the total genset fleet of Cameroon at 100,000 in 2016. This corresponded to a total genset installed capacity of about 1000 MW in 2016. The genset fleet is made up of 5% gasoline (petrol) and 95% diesel gensets, distributed in both grid-connected and off-grid areas. The less efficient gasoline gensets are mainly used by residential installations for service of small (often less than 5 kW) loads. Diesel gensets which can go up to 750 kW A are mainly used in commercial and industry installations. Electricity generation from backup gensets is estimated at over 1000 GWh annually [30–32].

Despite the prevailing issues in the Cameroon power sector, the nation with a GDP of 39.8 billion USD in 2020 and surface area of 475,442 km² is endowed with varied and abundant energy sources for electricity generation [23]. Theoretical electricity generation potential from hydropower is estimated at 23 GW [33]. Small hydropower potential is estimated at 792 MW [2]. With a national average irradiation of 4.9 kWh/m², solar photovoltaics (PV) technical generation potential is estimated at 10,105 TWh annually according to the International

Renewable Energy Agency (IRENA) [34]. Biomass electricity generation potential is estimated at 39761.4 TWh [2]. Onshore wind, though not very viable nationally, its technical potential at suitable areas at 30% capacity factor is estimated at 15.9 TWh/year [34]. Cameroon has historically been an exporter of crude oil since 1977. Natural gas exportation began recently in 2007 [2]. Proven reserves of natural gas are estimated at 4.8 trillion cubic feet. The country also has uranium, with reserves estimated at 24125 metric tons according to the Extractive Industries Transparency Initiative (ETI) [35]. Given the abundant and diverse energy resources, the nation shouldn't be grappling with inadequate generation.

3. Methodology and data

3.1. The LEAP/NEMO optimization framework

The Low Emissions Analysis Platform (LEAP) is an accounting and simulation model developed by the Stockholm Environmental Institute. Previously called Long-Range Energy Alternatives Planning, LEAP is a widely used tool for integrated energy planning, climate change mitigation and air pollution abatement policy analysis [36]. Optimization in LEAP is possible through a transparent interconnection with an optimization tool, where LEAP is used as a modelling interface. The choice of optimization tool is the Next Energy Modelling system for Optimization (NEMO), an open-source energy system deterministic optimization tool suitable for linear (LP) and mixed integer linear programming (MILP), especially for large problem sets. NEMO is built on Julia, a programming language originally developed by the Massachusetts Institute of Technology (MIT) [37]. In this study LEAP version: 2020.1.0.48 and NEMO version 1.60 was applied to assess the least-cost generation expansion pathways for the power system of Cameroon. From stated policies, the expansion of current power system is skewed towards large hydropower and thermal generation with minimal focus on grid-connected renewable energy sources.

The objective of NEMO model is to evaluate the least-cost generation expansion pathway required to adequately meet electricity demand and sought reserve margin, at minimum discounted total cost. This discounted cost constitutes: operation and maintenance (O&M) costs, capital/investment costs, feedstock fuel costs and externality costs of all generation plants. The model is subject to: feedstock fuel production, capacity and activity constraints. Feedstock fuel production must be greater than or equal to any exogenous or endogenous demand plus consumption by technology for that fuel in each time slice and year. For the capacity constraint, each technology capacity must be greater than or equal to its required activity in each demand segment. Lastly, the generation capacity of each technology is less than or equal to its annual availability in the activity constraint. It is also assumed that electricity transmission and distribution network capacity at each time slice and year will adequately transport the electricity demanded [36].

3.2. Scenario definitions

The scenarios in this study were developed using an adaptation of the scenario development methodology originally presented by Peter Schwartz [38]. Using the methodology, three critical factors or drivers influencing total electricity demand, generation mix and associated GHG emissions were identified. These are: transmission and distribution losses; demand side energy efficiency measures and emission reduction targets were identified. These drivers were labeled respectively as: power losses (PL), demand-side energy efficiency (DEE) and emission target (ET).

Two implementation levels; the Low and the High were assumed for each factor. The Low implementation level epitomizes the state's reluctance to fully implement her stated measures and energy sector masterplans. In contrast, the High implementation level represents a drastic and positive shift in enforcing energy sector measures and

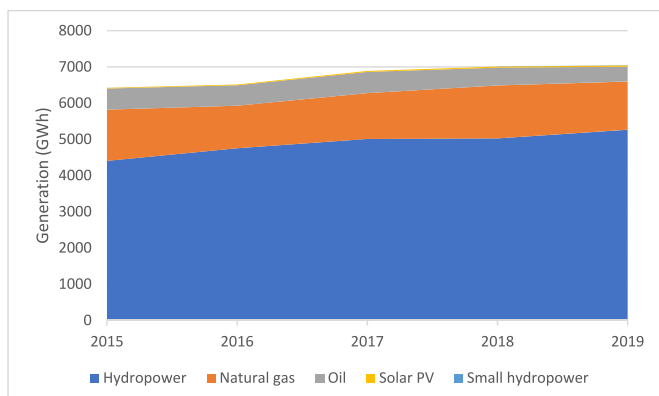


Fig. 1. Electricity generation by source in Cameroon.

masterplans by the state. The rationale for selection of two implementation levels come from the consideration of Median and Top scenarios in most government masterplans [2].

Thus, the resulting six scenarios, in addition to the BAU scenario were modelled as alternative least-costs electricity generation expansion pathways for Cameroon. The least-costs pathway in each scenario was developed by enabling LEAP/NEMO framework to endogenously decide on the generation technology type, size and dispatch rules [36]. The seven scenarios are presented in Fig. 2.

3.2.1. The business-as-usual (BAU) scenario

The BAU scenario represents the least-costs generation expansion pathway of Cameroon under the current trajectory and in stated masterplans. The current trajectory would see the exploitation of the nation's abundant hydropower potential, expansion of the thermal fleets and miniature emphasis on renewable energy. Addition of new hydropower capacity is based on planned projects such as the development of the: Nachtigal hydro (420 MW, ongoing and scheduled to be delivered in 2023), Song Dong (316 MW), Kikot 659 MW and the Grand Eweng (1800 MW), etc. as stated in the Energy Sector Development Project (PDSN) [2]. Thermal fleet expansion includes the rehabilitation of the Kribi natural gas plant to increase capacity, and conversion of older oil plants (such as the Limbe plant) to natural gas plants. The BAU scenario sees a slow development of renewable energy guided by stated measures in the Rural Electrification Master Plan (REMP) [5] and the NDCs [8].

The reserve margin is assumed to increase to 20% by 2045 from the presumed 5% in the base year to cope with the increasing variability in demand and ensure better grid reliability. Thus, the BAU scenario is the least-cost power system pathway calculated by the optimization model with no additional policy shifts or interventions. The results from the BAU scenario serves as a benchmark for assessing the alternative least-cost scenarios of the Cameroon power system under different policy considerations.

3.2.2. The power losses (PL) scenario

Power losses is among the leading cause of grid sub-optimal performance and low reliability, in both rural and urban areas of Cameroon [29]. Generally categorized into technical and non-technical losses, the former is attributed to line overloading and the general ageing grid infrastructure [29,39]. The latter are non-technical, largely attributed to illegal connections (theft) and nonpayment of accumulated bills by large customers (mainly the state) [21]. Power losses in Cameroon are normally divided into transmission and distribution losses, denoting the level at which it occurs. These losses have experienced a slow but steady growth over the years as presented in Fig. 3. In 2019, transmission and distribution losses respectively, were 6.9% and 32.03% according to

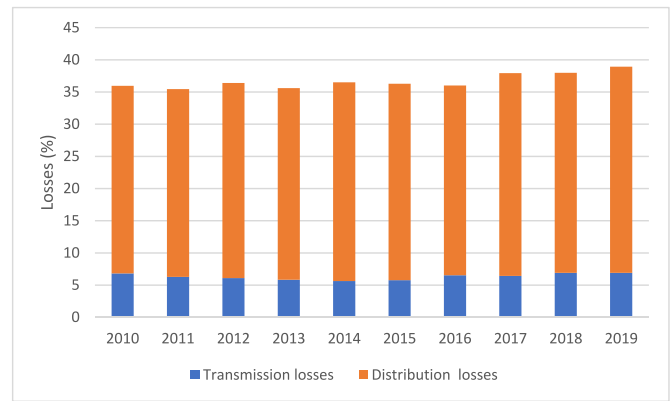


Fig. 3. Evolution of transmission and distribution losses in Cameroon [21, 26,27].

Ref. [21]. The combined network losses of 38.93% is way higher than the 2014 averages of 12% in sub-Saharan Africa average and 8.2% in world average according to the World Bank [23].

The state has earmarked several measures aimed at reducing power losses and improving the quality of service as highlighted in Refs. [29, 40,41]. Measures at the transmission level generally aim to: optimize electricity transport at the 225 kV and 90 kV lines by the installation of on-peak capacitors; and use of Flexible Alternating Current Transmission Systems (FACTS). Measures at distribution levels aim to improve distribution system efficiency through: phase balancing, capacitor placements at end of medium voltage (MV) feeders; and roll-out of smart-meters (prepaid) for billing systems [21]. Universal measures include installation of supervisory control and data acquisition (SCADA) system and deployment of infra-red cameras to detect network hotspots resulting from imperfect and ageing connections [26].

The power losses scenario in this study evaluates least-costs generation pathways resulting from implementing two levels of losses reductions policies in the network. In the Low PL scenario (PL1), power losses are interpolated to grow from 36.01% in 2016 to 38.93% in 2020, then assumed to linearly decrease to 20% in 2045. This aims to half power losses in Cameroon by 2045. In the High Transmission scenario (PL2), power losses are interpolated to grow from 36.01% in 2016 to 38.93% in 2020, then assumed to linearly decrease to 8% in 2045. This is in line with the 10.3% in 2045 average forecast of power losses in three African countries (Ghana, Sudan and Gabon) by Ref. [42].

3.2.3. The demand-side energy efficiency (DEE) scenario

Energy efficiency essentially entails minimizing the quantity of

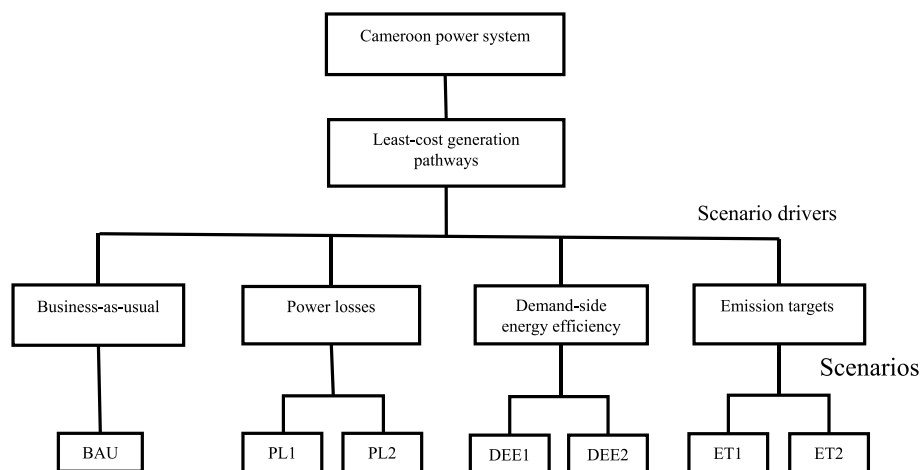


Fig. 2. Description of scenarios.

energy consumed for a specified, constant energy service [43]. The absence of electrical appliance standards or any consumer-oriented demand side management policy in Cameroon have intensified the proliferation of inefficient (mostly second-hand) electrical appliances. The Electricity Regulatory Agency, ARSEL in 2014 conducted the National Energy Efficiency Policy (NEEP) study [40]. The NEEP study identified and evaluated possible demand-side energy efficiency measures and associated energy savings, compared to the business-as usual scenario. Appropriate technological improvements and accompanying policies were exclusively presented in three sectors: residential and end-user; tertiary and buildings; and industry sectors [40,41].

From the NEEP, industry sector measures include: optimizing waiting phases in production lines, regulation of energy intensive equipment such as compressors and ditching the use of obsolete technologies. In addition to these measures, potential electricity savings of between 20% and 30% are achievable depending on how quickly co-generation technologies are adopted. Electricity-saving measures in the tertiary and buildings sector are primarily concerned with lighting, air conditioning system and the development of a building energy quality code for new buildings. From these measures, up to 30% electricity savings can be achieved in the tertiary and buildings sector. Residential and end-user sector savings are mostly in household appliance standardization and use of more energy efficient appliances (bulbs, fridges, air conditioners, etc.), in addition to building developing building energy codes. Implementing these measures could lead to up to 30% potential energy savings.

The DEE scenarios evaluate the impact of two levels of demand-side energy efficiency policy implementations on the least-cost electricity generation expansion pathway of Cameroon. The Low demand-side energy efficiency (DEE1) scenario assumes electricity demand savings of 15% by 2045 from implementing lesser capital-intensive measures. The High demand-side energy efficiency (DEE2) scenario assumes energy savings of 30% by 2045 from implementing more capital-intensive measures. Energy savings under both scenarios are calculated from the NEEP.

3.2.4. The emission target (ET) scenario

Cameroon in 2015 enacted her Nationally Determined Contributions (NDCs) [8]. The NDC aim to attain a 32% CO₂ emission reduction in 2035, compared to the business-as-usual forecast of 104 MTCO₂e. Emission reduction is targeted from the agriculture, forestry, waste, transport and industry sectors. The power sector is expected to contribute 4.5% (4684 MTCO₂e) to the 32% planned CO₂ reduction. Stated power sector measures in the NDCs, in addition to the renewable energy projects in the Rural Electrification Masterplan include attaining a 25% share of renewable energy (6% solar PV, 11% small hydro, 7% biomass and 1% wind) in the generation mix by 2035 [5,8]. However, the 2020 renewable energy share in the grid is <0.5%, far less compared to the anticipated 3% renewable energy target in NDCs. Even with the efforts by the state, the nation is however lagging in her NDC commitments.

The emission target (ET) scenarios in this study assess the impact of setting emission targets on the least-cost generation expansion pathway of Cameroon. However, the energy generation structure is largely dependent on hydropower (75%), with the rest from natural gas 19% and oil 6% [21]. To significant assess the impact of emission targets, and in line with stated policies, all existing oil plants are expected to be retired or upgraded to natural gas plants in the near future. In these scenarios, total capacity of oil plants is assumed to reduce by 50% in 2035 and by 100% in 2045.

In the Low ET scenario (ET1), a 50% CO₂ emission reduction target is imposed on the power sector from 2021. ET1 aims to meetup with the stated power sector emission reduction contribution targets in the total NDCs. The High ET scenario (ET2) imposes a 100% emission reduction target from 2021. ET2 aims to examine an optimal path for the transition to a zero-carbon power system in Cameroon.

3.3. The Cameroon LEAP model

The bottom-up modelling methodology was used to developed the Cameroon power generation model in LEAP. The base year choice of 2016 was chosen due to the availability of reliable data, and commencement of most energy sector plans. The choice of base year also enabled the validation of model results with actual statistics from subsequent years (2017–2019). A real discount rate of 8% was adopted in this study, which is analogous to the 10% used in the West Africa Power Pool (WAPP), though Cameroon is in The Central African Power Pool (CAPP) [44].

Though a lower middle-income country at present, Cameroon's GDP in this study is assumed to grow from 32.64 billion USD in 2016 at a rate of 3.34% between 2016 and 2020 and 7% from 2021 to 2045 (GDP of 101.79 billion USD by 2045). The high GDP growth rate is necessary if the nation is to achieve her goal of being a newly industrialized and upper middle-income economy as stated in Ref. [1]. During this same period, population increases from 23.9 million in 2016 at a rate of 2.6% annually through to 2025 and at 2.3% annually through to 2045 to 47.4 million in 2045. Electrification rates are assumed to reach 100% by 2030 for urban areas and 70% by 2045 in rural areas [2].

Total electricity demand was 6098 GWh in 2016 [27]. Demand is projected to grow at an average annual rate of 6.48%. This demand growth rate is similar to official rate of 6.7% under the median scenario in the 2014 PDSEN, a masterplan published by the Ministry of Water and Energy (MINEE) [45]. This high growth rate accounts for increase in electrification rates, fulfilment of suppressed demand and an increase in urbanization rate, the latter projected to grow to 68.8% by 2045 [46].

Total dependent installed capacity in 2016 was 1322.5 MW. This consisted of large hydropower (55.4%), natural gas (22.6%), oil (21.2%) and renewables (0.7%) [25,27]. Total generation in 2016 was 6513 GWh. The trend is presented in Table 1 [27]. Renewable energy technologies considered in this study are: small hydropower, solar PV, wind and biomass, with only solar PV (9 MW) and small hydropower (0.3 MW) contributing in 2016.

Generating plants were aggregated based on conversion technology. For example, all large hydropower plants were combined and modelled as one hydro generator. The reserve margin was also assumed to increase from 5% to 20% in order to meet the excess capacity required to meet future peak demand, a lagging factor at present. Transmission and distribution losses in 2016 were 36.01% [27]. The technical characteristics of generation technologies are presented in Table 2.

Generation technology cost data were adopted from the Annual Technology Baseline of the National Renewable Energy Laboratory (NREL) [48] and the Renewable Energy Costs 2019 of IRENA [49]. These costs consist of capital costs, variable and fixed operation and maintenance costs of future generation plants. Capital costs of conventional technologies were assumed to be constant over the study period in contrast to renewables which decrease with time. These costs are presented in Table 3.

The volatility of fossil fuel prices makes them difficult to accurately project. However, reliable fossil fuel price forecasts were adopted from the Annual Energy Outlook 2021 [50] and World Energy Model 2020 [51], both by the IEA. Biomass feedstocks are very heterogeneous, with costs generally dependent on the type, size and moisture content. Owing to their abundance, biomass feedstock is largely untraded in Cameroon, with possible costs only incurred from handling and transportation. However, an average feedstock price of 15\$/ton in 2016 was assumed, in spite the wide variety of feedstock present in Cameroon. This cost is assumed to linearly grow to 30\$/ton by 2045 [52]. These prices and forecast are shown in Table 4. Other renewable energy sources are considered cost-free.

3.4. Sensitivity analysis

The technology type added, total capacity, and eventual generation

Table 1
Installed capacity and generation in Cameroon [21,25–27].

Type	2016					2020				
	Total	% Share				Total	% Share			
		Hydro	Gas	Oil	renewables		Hydro	Gas	Oil	renewables
Capacity	1322.5 MW	55.4	22.6	21.2	0.7	1541.4 MW	61	20	18	1.0
Generation	6513 GWh	73.0	18.0	8.7	0.3	7043 GWh	74.7	18.9	6	0.4

Table 2
Technical characteristics of power generation technologies [25,47].

Technology	Capacity factor (%)	Efficiency (%)
Large hydropower	60	100
Small hydropower	60	100
Natural gas	80	30
New natural gas	85	48
Existing oil	80	30
New oil	80	35
Solar PV	25	100
Wind	35	100
Biomass	50	35

In this study, a constant discount rate of 8% is assumed. The effect of inflation rate on energy sector investments were ignored. Further, the electricity sales revenue and salvage value were ignored given the same demand, growth and reserve margin were considered in all the scenarios. Also, the investment costs necessary to implement the measures in the Power Losses, Demand-side Energy Efficiency and Emission Target scenarios are beyond the scope of this study, so wasn't considered. The costs considered in this study are: the capital costs, operation and maintenance (both fixed and variable O&M) costs, feedstock fuel costs and CO₂ externality costs.

Table 3
Technology cost data considered in the Cameroon LEAP/NEMO model.

Technology	Capital costs (\$/kW)		Fixed O&M costs (\$/kW-yr)		Variable O&M costs (\$/MWh)	
	2016	2045	2016	2045	2016	2045
Large hydro	1750	1750	22	22	8	8
Small hydro	2700	2700	28	28	12	12
Natural gas	1088	936	13	13	2	2
Oil	1088	936	13	13	2	2
Solar PV	2052	723	18	11	–	–
Wind	1751	1296	43	43	5	5
Biomass	3000	3000	42	42	5	5
MSW	6400	5800	250	220	25	20

Table 4
Fuel prices considered in the Cameroon LEAP/NEMO model.

Fuel type	Prices						
	2016	2020	2025	2030	2035	2040	2045
Natural gas (\$/mBtu)	3.2	3	3.1	3.5	3.8	4.1	4.6
Crude oil (\$/barrel)	40.7	40.4	66.0	88.9	110.4	131.9	157.9
Biomass (\$/ton)	15.0	16.5	18.6	21.0	23.6	26.6	30.0

mix depends on model assumptions. Electricity demand is largely dependent on several factors such as population, GDP growth, discount rates and energy efficiency policies. The least-costs generation pathway depends on technology costs and sector targets. Sensitivity analysis is often used to evaluate the accuracy and robustness of model results.

In this study, capital and O&M costs for thermal plants are assumed constant, while they constantly decline for renewables. However, prices of feedstock, particularly fossil fuels utilized by thermal plants are very volatile, often fluctuating in world markets. Furthermore, the assumed

average population growth rate over the model horizon doesn't necessarily reflect recent global demographic trends. Thus, a sensitivity analysis on population growth and feedstock fuel costs was performed in this study. A $\pm 10\%$ uncertainty was assumed for each factor. Also, an assessment of discount rate change was also performed.

Scenario combination is performed to evaluate the impact of policy mixing on the least-cost generation pathways. Two scenarios were considered:

- (i) Scenario 1: combination of PL1 and DEE1,
- (ii) Scenario 2: combination of PL1, DEE1 and ET2.

4. Results and analysis

The least-cost generation system of Cameroon resulting from the data and assumptions in Section III over the 2016–2045 study period are presented in this section.

4.1. Electricity demand

The electricity demand of Cameroon grows from 5.43 TW in 2016 to 33.3 TW in 2045. This designates an average of 6.5 annually as shown in Fig. 4. The demand growth in this study corresponds to and official projections, the latter projecting an average growth rate of 6.7% [21, 45]. This demand growth is shown in Fig. 4.

4.2. Installed capacity under optimization scenarios

The installed capacities are derived by enabling LEAP/NEMO decide on the type and quantity of technology to be added to meet the electricity demand and 20% reserve margin of Cameroon.

4.2.1. Installed capacity under the BAU scenario

The total installed capacity under this scenario represents no changes in planned state policies over the course of the study. Under the BAU scenario, installed capacity increase from 1.3 GW in 2016 to 13.4 GW in 2045 as shown in Fig. 5.

From Fig. 5, the high cost of fossil fuels coupled with high operation and maintenance costs cause a decrease in thermal generation. This is

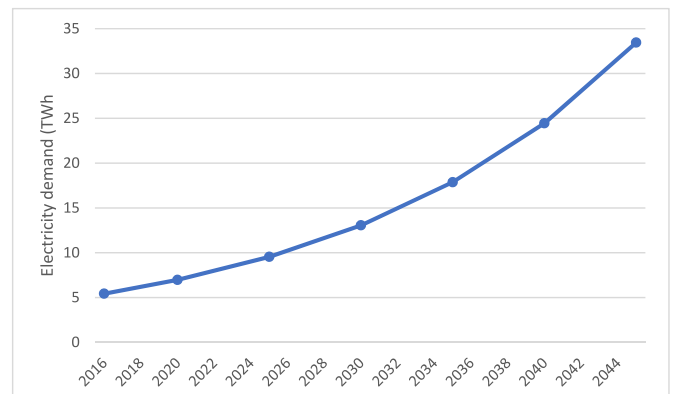


Fig. 4. Electricity demand of Cameroon between 2016 and 2045.

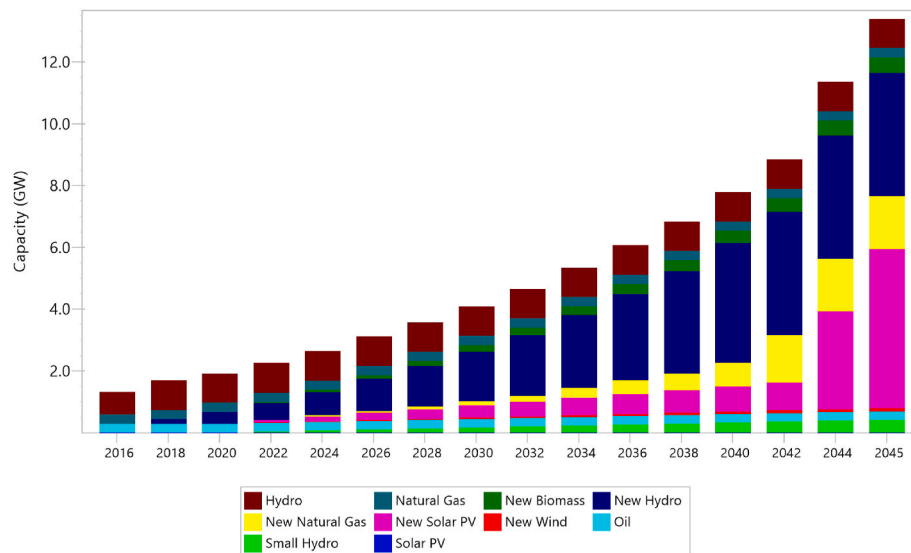


Fig. 5. Installed capacity by technology under the BAU scenario.

evident by the 27% decrease in capacity share of thermal plants in 45% compared to the base year. Also, no new capacity is added for oil plants under this scenario emphasizing the high lifecycle costs of oil generation plants.

Energy generation under the scenario reaches 54.8 TWh in 2045. Hydropower share drops slightly to 70% in 2045, compared to the 73% in the base year. Renewable energy (wind- 0.6%, solar PV- 20.7%, small hydro – 2% and biomass- 0.1%) makes up over 24% of energy generation in 2045. Thermal generation which consists of only natural gas, making up 6%. The trend of energy generation is presented in Fig. 6.

4.2.2. Installed capacity under the power losses scenario

In the Power Losses scenarios, there is a decrease in total capacity compared to the BAU scenario. Installed capacity increases from 1.3 GW in 2016, to 8.4 GW and to 7.5 GW under PL1 and PL2 respectively. This is presented in Fig. 7.

From Fig. 7, the willingness to reduce network losses from 38.93% in 2020 to 20% in 2045 (PL1) would result in avoided installed capacity of up to 5 GW (avoided capacity in white) in 2045 compared to the BAU scenario. This capacity savings corresponds to more than the base year

(2015) and current (2020) installed capacities of 1.3 GW and 1.5 GW respectively. However, avoided installed capacity of up to 5.9 GW in 2045 compared to the BAU scenario is achievable depending on commitments to improving transmission and distribution efficiency. This projected 5.9 GW capacity savings is more than the total anticipated hydropower capacity in the Energy Sector Development Project (PDSEN) of the Cameroon's Ministry of Energy and Water Resources [5].

Under the power losses scenario, the required energy generation to meet demand and reserve margin in 2045 is 41.8 TWh and 36.4 TWh under PL1 and PL2 respectively. This corresponds to savings of 13.0 TWh and 18.4 TWh respectively, compared to the BAU Optimization scenario. The energy generation trend is presented in Fig. 8.

4.2.3. Installed capacity under the demand-side energy efficiency scenario

Total installed capacity in 2045 under DEE1 and DEE2 reaches 9.9 GW and 7.8 GW respectively, from the base year value of 1.3 GW these results are presented in Fig. 9.

From Fig. 9, the introduction of demand side energy efficiency measures as stated in the National Energy Efficiency Policy (NEEP) study has significant impacts on the total installed capacity. Application of

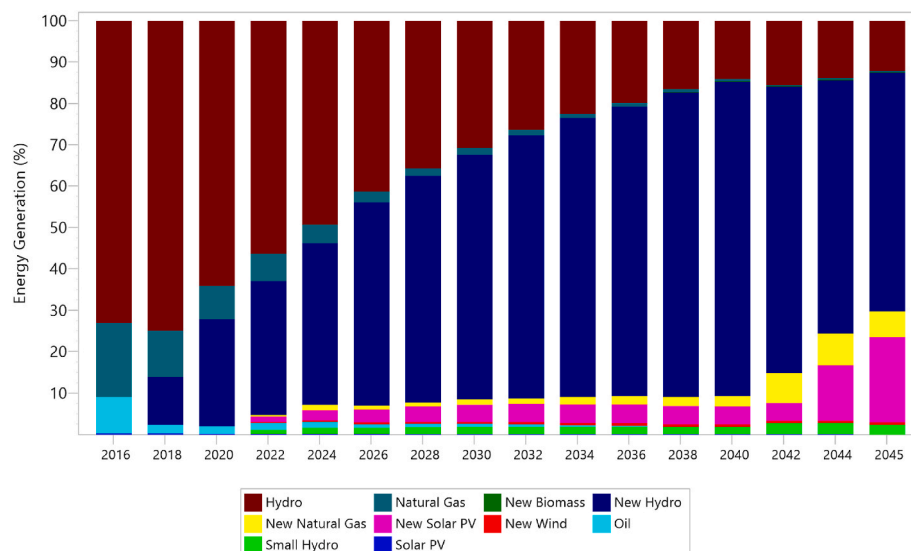


Fig. 6. Energy generation by technology share under the BAU scenario.

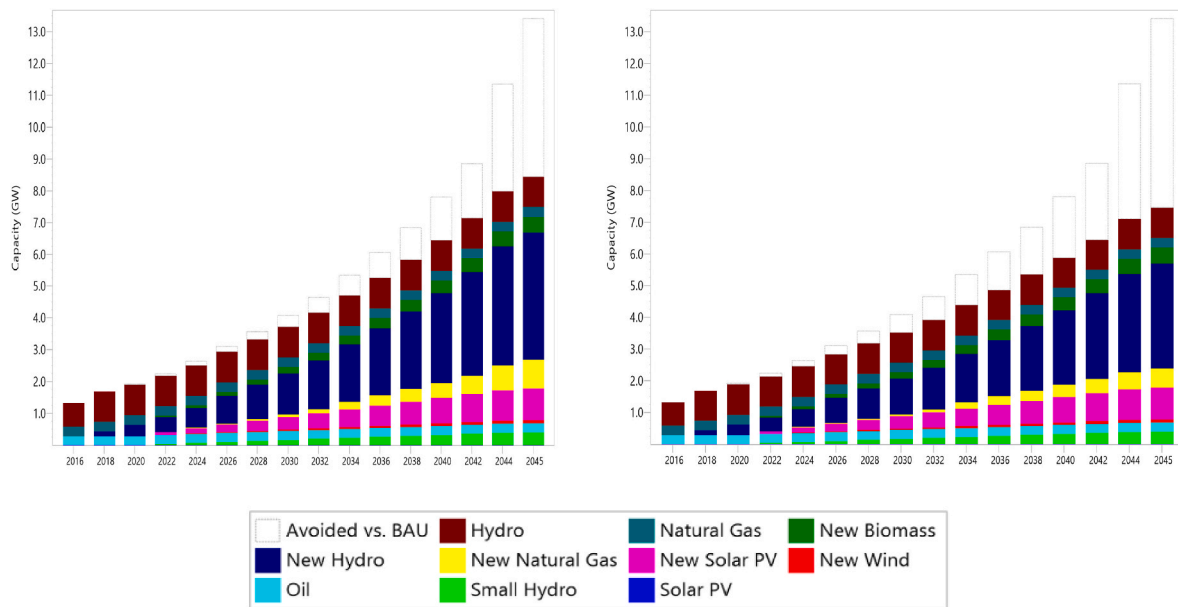


Fig. 7. Installed capacity by technology under the PL1 and PL2 scenarios.

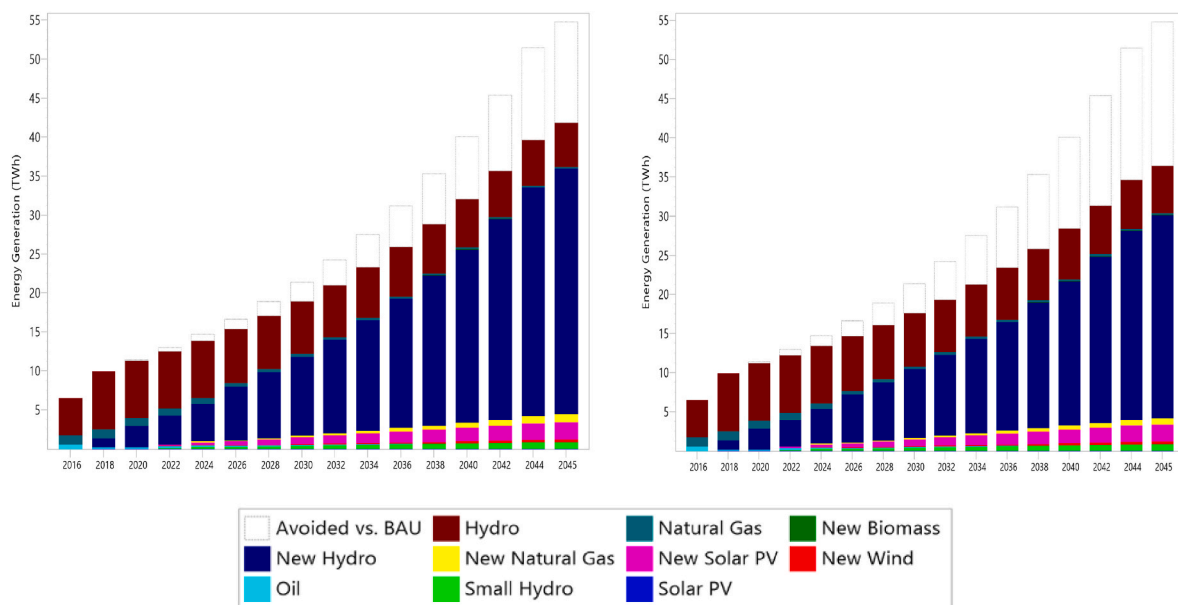


Fig. 8. Energy generation by technology under PL1 and PL2 scenarios.

non-intensive energy efficiency measures, like in DEE1 results in 3.5 GW in 2045 of avoided installed capacity, almost equivalent to the current (2020) total installed capacity. However, more ambitious implementation of energy efficiency measures as in EE2 would result to capacity savings of up to 5.6 GW. The trend is presented in Fig. 9.

The required energy generation in 2045 is 46.5 TWh and 38.4 TWh under DEE1 and DEE2 respectively. This corresponds to avoided generation of 8.3 TWh and 16.4 TWh compared to the 54.8 TWh under the BAU scenario. This is shown in Fig. 10.

4.2.4. Installed capacity under the emission target scenario

The introduction of emission reduction targets in Cameroon's power sector leads to mixed results. Introducing an emission target of 50% under the ET1 scenario results in installed capacity of 14.5 GW in 2045, an 8% increase compared to the BAU scenario. This reiterates the small contribution of fossil fuel in the total generation mix and the nation's

trajectory to developing more of hydropower and renewables-cleaner fuels.

However, introducing a 100% emission reduction target as under the ET2 scenario results in high increase in installed capacity. Installed capacity under ET2 reaches 18.0 GW in 2045. A 24% and 34% increase respectively, compared to ET1 and BAU scenarios. This increase is mainly to account for the increased share of renewable energy technologies. The installed capacities are presented in Fig. 11.

The energy generated under both ET1 and ET2 is at 54.8 TWh in 2045, the same as in the BAU scenario. However, ET1 and ET2 scenarios show variations in generation mix, an increase in shares of renewables unlike in the BAU scenario. This is presented in Fig. 12.

The difference between ET1 and ET2 is to the influence of renewables, which need higher capacities needing to substitute the early retirement of thermal plants.

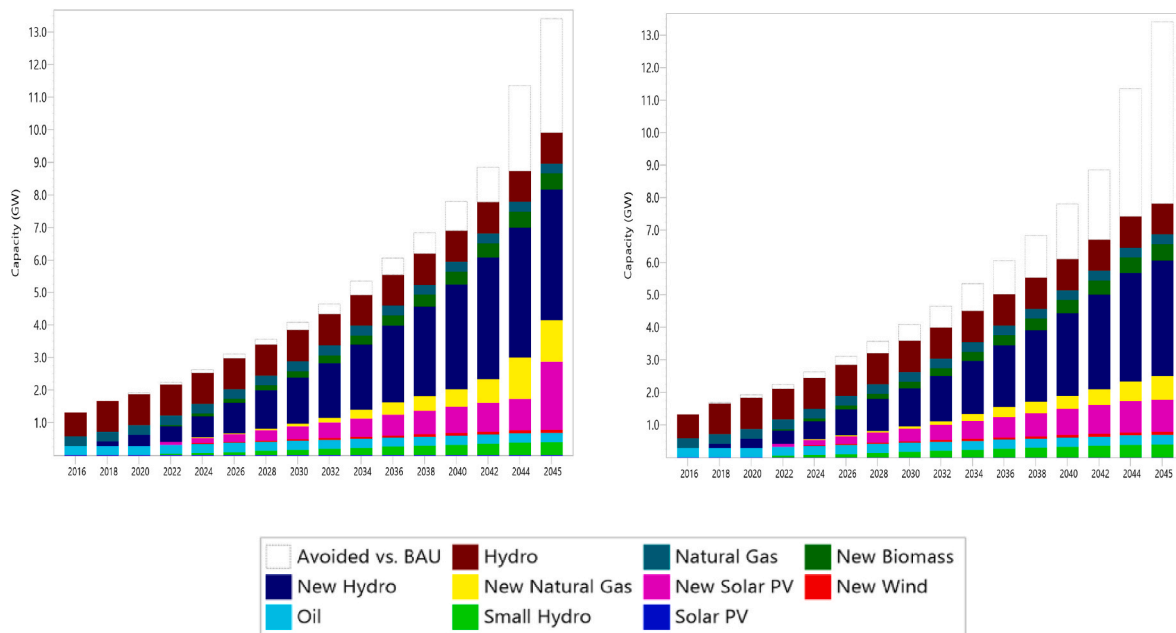


Fig. 9. Installed capacity by technology under DEE1 and DEE2 scenarios.

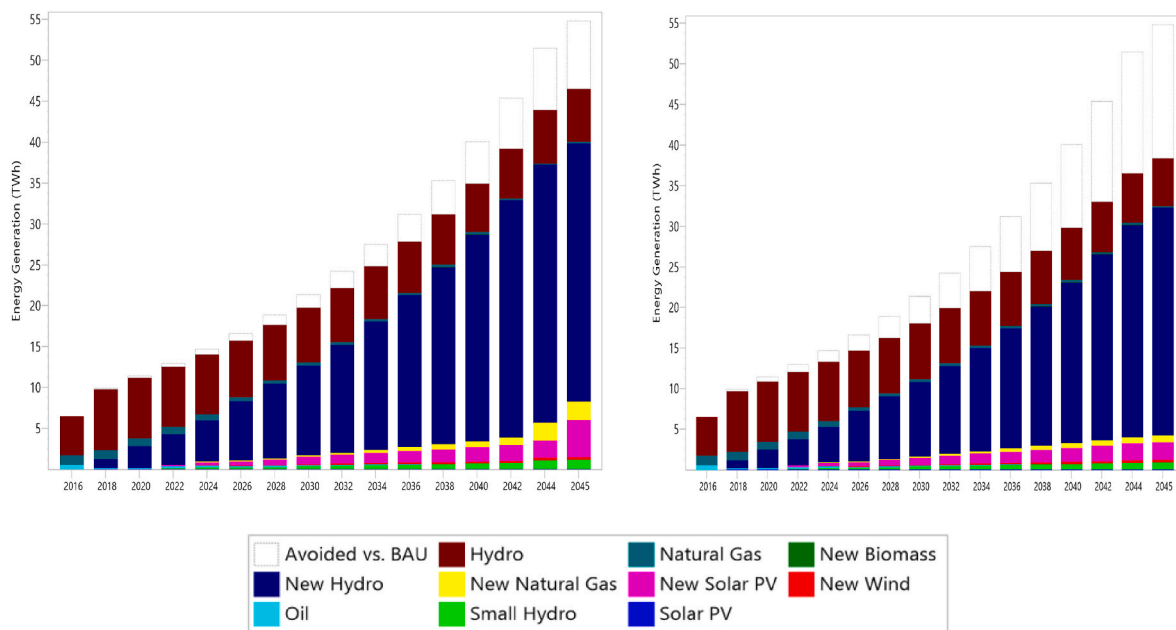


Fig. 10. Energy generation by technology under the DEE1 and DEE2 scenarios.

4.3. Economic analysis

The LEAP/NEMO optimization model assesses the least-costs energy development pathway. The least-costs are usually expressed financially in the form of a Cost-benefit analysis, and the total in terms of the Net-Present-Value (NPV). The NPVs expressed in this study are cumulative cost-benefits over the study duration, and expressed in 2016 US dollars.

The cumulative NPV under the BAU scenario in 2016 US dollars, discounted at a rate of 8% reaches 2547 million USD in 2045. This cost constitutes 57% capital costs, 35% O&M and 8% fuel costs. The low fossil fuel costs signify the small contribution of thermal plants to the energy generation of Cameroon.

The cumulative NPVs by cost type of the alternative scenarios compared to the BAU scenario is presented in Table 5. A negative NPV

indicates lesser costs with added benefits while a positive NPV indicates a higher costs and lesser benefits, all compared to the BAU scenario.

From Table 5, all alternative scenarios experience cost-savings as indicated by their negative cumulative NPVs apart from the ET2 scenario. The positive NPV in the ET2 scenario compared to the BAU scenario is largely due to renewables. Though substantial savings are attained from fuel costs, it is not enough to offset the high capital and O&M costs of renewables, in addition to hydropower under the ET2 scenario (Fig. 5).

Furthermore, the implementation of power loss reduction measures could result in cost savings of up to 603 million USD (PL2) if power network losses could be reduced to the world average of about 8% [23]. Though the implementation costs of the stated measures wasn't included in this study, reducing power losses hold the most promise of

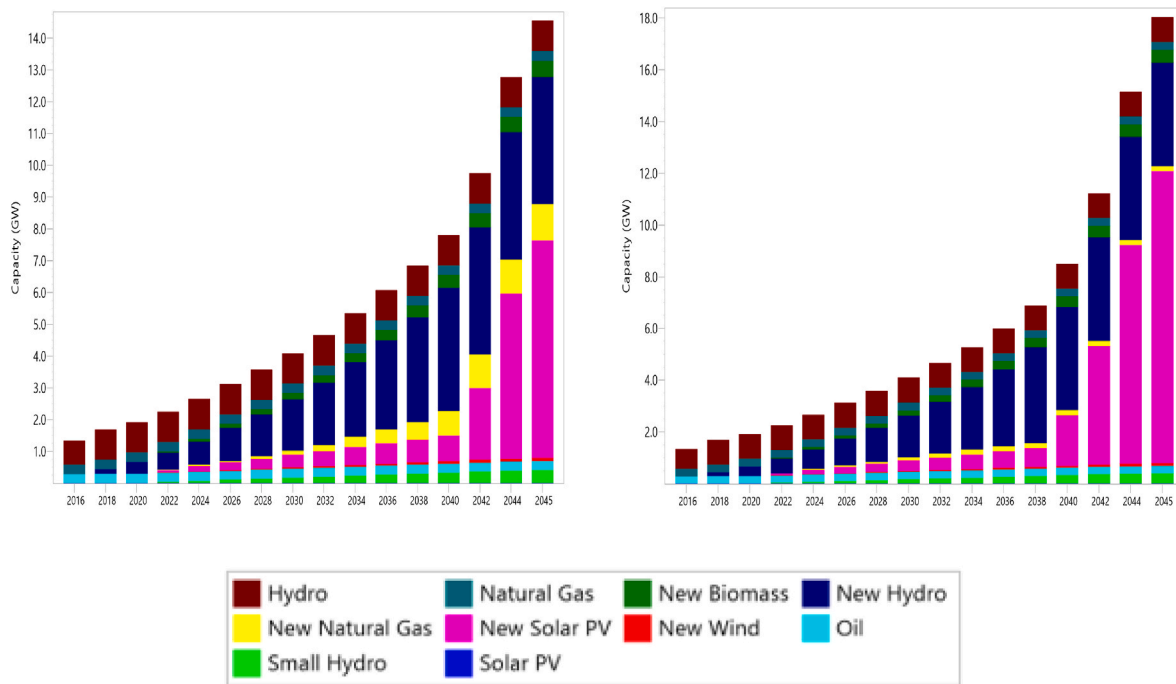


Fig. 11. Installed capacity by technology under ET1 and ET2 scenarios.

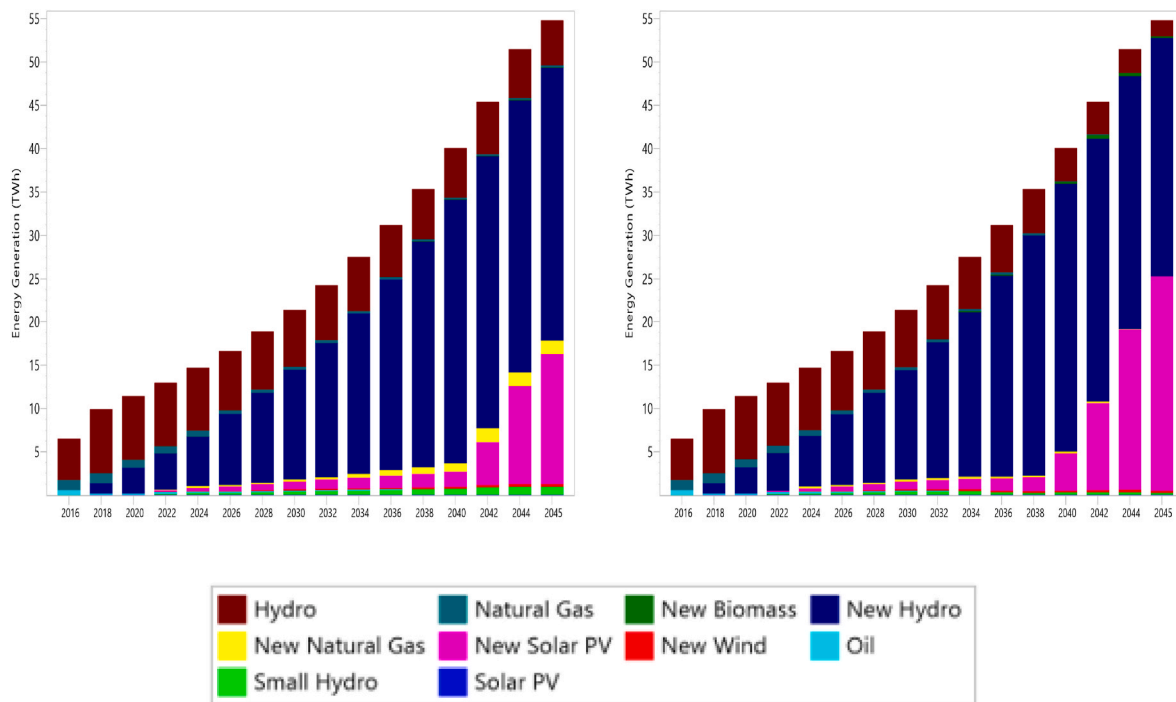


Fig. 12. Energy generation by technology under ET1 and ET2 scenarios.

net benefits for the state, given the 603 million USD represents over 57% of cumulative costs under the BAU scenario.

Also, Energy efficiency measures for the demand side energy efficiency and emission targets policies from Table 5 also show positive benefits for the Cameroon power system if these actions are taken. However, an NPV of 7 million USD in ET1 indicates the implementation of a 50% emission target reduction would have little financial impact on the least-costs generation pathway of Cameroon. This is due to the already low contribution of fossil fuels in any optimization based on

cost-benefits. Thus, the state should aim achieving above 50% emission reduction if it wants to significantly decarbonized her power system. Furthermore, the state should prioritize the development of renewables rather than the planned expansion of the thermal fleet.

The Paris Agreement and global movement to transitioning to a Green Economy have seen an increased interest in national carbon pricing and carbon trading. Despite the many financial benefits, high potentials and discussions in sub-Saharan Africa, strategies to implement this in the region has been futile. Only South Africa of the 54

Table 5

Cumulative NPV of alternative scenarios at 8% discount rate relative to the BAU scenario (in million 2016 USD).

Cost Category	Power Losses		Demand-side Energy Efficiency		Emission Targets	
	PL1	PL2	DEE1	DEE2	ET1	ET2
Capital	-252	-366	-166	-329	35	133
O&M	-121	-184	-76	-164	-5	-17
Feedstock Fuel	-48	-54	-36	-52	-23	-15
Cumulative NPV	-420	-603	-278	-545	7	101
Externality	-27	-30	-20	-29	-16	-41
Cumulative NPV with externalities	-447	-633	-298	-573	-10	61

Negative values indicate lesser costs and more benefits while Positive values indicate more cost with less benefits.

countries in Africa currently implement carbon taxes. Though carbon taxes are under consideration in Senegal and Cote d'Ivoire, implementation has met with difficulties [53]. This has been attributed to several factors including lack of relevant studies and paucity of data [54]. This paper aims contribute to the discussion, by quantifying in financial terms the potential revenue if carbon pricing were instituted in the Cameroon power system. Thus, a carbon tax of 15 USD/metric tonne of CO₂ was introduced in 2025. This externality cost is linearly interpolated to increase to \$50/metric tonne of CO₂ in 2045.

Under the BAU scenario, the cumulative externality costs discounted at 8% and expressed in 2016 USD is 58 million USD. The influence of externality costs on the NPVs is also shown in Table 5. From Table 5, its observed that carbon tax has no significant influence on the direction of cost-benefits. This is due to the low carbon low contribution of thermal plants in Cameroon's generation mix.

However, the influenced of the carbon tax is seen in ET2 from Fig. 12. The carbon tax of 58 million USD reduces the comparative NPV of the ET2 scenario by 40%. Moreover, the revenue for carbon pricing could help fund the development of renewable energy sources.

4.4. Environmental analysis

The LEAP model performs its environmental assessment based on the Global Warming Potential (GWP) values in the Fifth Assessment Report (AR5) published by the Intergovernmental Panel on Climate Change (IPCC) [55]. The GWP at point of emissions in this study and expressed in million MTCO₂e.

Under the BAU scenario, GHG emissions increase to 1.6 MTCO₂e in 2045, from 1.3 MTCO₂e in 2016. This is mainly from natural gas plants - 99.9%, and about 1% from biomass generation plants. Over the study period, total GHG emissions cumulate to 22.7 MTCO₂e. The GHG emissions of other scenarios compared to the BAU scenario is presented in Table 6.

From Table 4, the decrease in emissions despite the increase in demand and associated generation is attributed to retirements of, and fuel switching from oil to natural gas plants. Another reason is the assumption of combined-cycle power plants (CCPP) for new natural gas

Table 6

Cumulative GHG emissions under all scenarios.

Scenario	Emissions in 2045 (MTCO ₂ e)	Cumulative emissions (MTCO ₂ e)	Cumulative emissions savings compared to BAU scenario (%)
PL1	0.55	16.98	25
PL2	0.46	16.20	29
DEE1	1.06	18.47	19
DEE2	0.49	16.52	27
ET1	0.80	19.39	15
ET2	0.003	13.73	40

installations which are 50% less emitting instead of the mostly conventional presently in use in Cameroon [56].

Furthermore, its observed that the 100% emissions reduction objective under ET2 is not achieved, despite the retirement of all fossil fuels plants in 2045. The 0.003 MTCO₂e in 2045 is solely a result of biomass generation, which contributed about 0.3% (0.2 TWh) to the total generation mix (Fig. 12). this is contrary to the "waste and sequestration" arguments often postulated in literature [57]. Thus, biomass still contributes to GHG emissions at point of emission. If Cameroon were to achieved a truly carbon-free power system, the state should consider replacing biomass targets with solar PV, an abundant resource in the nation, given onshore wind is not very viable.

4.5. Sensitivity results

From the sensitivity analysis in Table 7, its observed that uncertainty in population growth rate have the expected impact on the least-cost generation pathways, i.e., decrease or increase leads to corresponding savings or increase in installed capacity, cumulative costs and GHG emissions.

Lower fuel costs have negligible effect on capacity and cumulative costs, although it leads to cumulative GHG emissions, given it favors fossil-fuel generation. The trend is similar for capacity and cumulative costs under higher fuel costs. However, higher fuel costs have an immense impact on emission savings compared to the corresponding impacts of low fuel costs. This indicates that higher fuel costs favor a greater transition to cleaner-fuels.

Furthermore, a lower discount rate (4%) leads to an increase in installed capacity and cumulative costs, despite cumulative GHG emissions savings as observed in Table 7. The trend is mostly reversed under

Table 7

Change in scenarios results (%).

Change in total installed capacity in 2045							
	BAU	PL1	PL2	DEE1	DEE2	ET1	ET2
Low population growth rate	17	8	8	14	8	23	18
High population growth rate	-19	-17	-8	-19	-9	-10	-18
Low fuel costs	2	0	0	0	0	10	0
High fuel costs	-3	0	0	-2	0	5	0
Low discount rate (4%)	-13	-2	0	-10	0	-4	-2
High discount rate (12%)	9	0	1	5	0	14	-1
Change in cumulative costs							
	BAU	PL1	PL2	DEE1	DEE2	ET1	ET2
Low population growth rate	8	7	7	7	7	8	9
High population growth rate	-7	-6	-6	-6	-6	-7	-8
Low fuel costs	0	0	0	0	0	1	0
High fuel costs	-3	-2	-2	-2	-2	-2	-1
Low discount rate (4%)	-57	-57	-57	-57	-57	-57	-56
High discount rate (12%)	134	134	135	134	135	136	134
Change in cumulative GHG emissions							
	BAU	PL1	PL2	DEE1	DEE2	ET1	ET2
Low population growth rate	-14	-4	-2	-8	-2	1	-1
High population growth rate	15	7	3	12	4	35	1
Low fuel costs	8	7	7	6	7	25	4
High fuel costs	-6	-4	-4	-5	-4	10	-2
Low discount rate (4%)	-45	-44	-46	468	-45	-36	-41
High discount rate (12%)	63	69	65	66	65	68	7

a higher discount rate of 12%. This indicates that lower discount rates favor uptake of generation technologies with higher capital costs and lower future expenditures (fuel and O&M costs) i.e., renewables. In contrast, higher discount rates favor generation technologies with lower capital costs but higher future expenditures i.e., fossil fuel plants. This reiterates the impact of discount rates on NPV calculations (upfront and future values of cash flows), and on resulting least-cost generation expansion pathway.

The combination of sustainable policies in Cameroon power system leads to greater benefits as shown in Table 8. We observed capacity and costs savings respectively, of up to 46% and 25% in Scenario 1, which is similar to Scenario 2. This indicates, the addition of emission targets has little impact on capacity and cumulative costs, compared to implementing just power loss reduction policies and demand-side energy efficiency measures. However, adding emission targets to power losses policies and demand-side energy efficiency measures would lead to more GHG emissions savings.

5. Discussion

This paper evaluates alternative least-cost generation pathways under three key factors driving power sector expansion in Cameroon. The strengths and novelty of this work is that it is the first of its kind being done in the nation and it incorporates both off and on grid generation. For the four questions facing the electricity sector and low-carbon energy transition of Cameroon as posed in the introduction, we obtain the following insides:

- Cameroon's generation mix in the BAU in 2045 reaches 13.4 GW. The generation system evolves towards a low-carbon power system, with the share of fossil fuels incrementally being substituted by renewable energy sources. This uptake which would be at cumulative costs of 2547 million USD by 2045 is as a result of the government's emphasis on renewable energy technologies. Cumulative GHG emissions over the study period cumulate to 22.7 MTCO_{2e} by 2045.
- Furthermore, the policies targeting power losses reduction are the most effective, compared to emission targets and demand side energy efficiency policies. Up to 5.9 MW and 603 million USD of installed capacity and cumulative costs can be avoided just by reducing power losses.
- Carbon taxing and emission trading is possible in the generation system of Cameroon. However, the carbon revenue can only be substantial at intensive emission policies (such as zero-carbon target) or at very high externality costs. Nevertheless, carbon taxing would result in installed capacity and cumulative costs compared to the BAU, although it holds the most decarbonization potential. For example, a zero-carbon system by 2045 compared to the BAU scenario would require 24% more installed capacity and 40% more in cumulative costs but 40% less GHG emissions.
- The future generation mix of Cameroon if a zero-carbon power system is sought would constitute entirely of renewables as presented in the ET2 scenario. Installed and energy generation capacity under ET2 respectively reaches 18.0 GW and 54.8 TWh in 2045. The energy generation mix constitutes: hydropower (53.4%), solar PV (45.3%), small hydro (0.4%), wind (0.6%) and biomass (0.3%).

However, the study has some limitations. Firstly, total genset

capacity, estimated at 1000 MW in 2016 (equivalent to about 76% of recognized installed capacity), and generating over 1000 GWh (equivalent to about 14% of recognized generation) wasn't considered in the study [21,30]. This is due to its nonrecognition in official policy discussions/masterplans, and lack of data on the annual generation profile given the spontaneous nature of power outages, amongst others. With increasing grid unreliability in the nation, the market deployment of backup gensets in Cameroon is only likely to improve. According to Ref. [31], Africa diesel genset deployment is projected to grow at annual rate (CAGR) of about 5% between 2022 and 2027. Infrastructural development projects such as: improving health care access, telecommunication network expansion (antennas) and increasing hospitality services are the primary drivers for this growth. Thus, the study results do not entirely reflect Cameroon's real total capacity growth, costs GHG emissions, nor the least-costs expansion pathway [58,59].

Furthermore, the study does not consider implementation costs of policies assumed in the alternative scenarios due to data unavailability. This would have enabled the quantification of the real costs associated with each pathway. The paucity of metadata is the motivation for U4RIA which aim to make data and standards of energy models more "ubiquitous, Retrievable, Repeatable, Re-constructible, Reproducible, Interoperable and Auditable, (u4RIA)". the goal of U4RIA is to address the common challenges in modelling in developing countries such as data paucity or incomplete metadata. This hopes to ensure practical goals in energy modelling for policy support [60].

The study utilized the bottom-up modelling approach and cost optimization framework provided in LEAP/NEMO to develop the scenarios. Cost optimization models often assume perfect foresight that rely on a number of parametric and structural assumptions in evaluating scenarios. However, these assumptions such as: future costs, discount rates, technology availability, demand growth, 100% grid capability, and service demands, etc. considered in this study are subject to variable degrees of uncertainty, which are often glossed over [61,62]. Cost-optimization models have been shown to consider narrow assumptions, have systematic biases, value-laden, fragile, and lead to irreproducible scenarios [63,64]. Evidence from literature indicates that these models do not adequately capture the structural changes in real-world transitions. An example is the study by Ref. [63], which obtained deviations between 9 and 23% on post modelling of the United Kingdom's (UK) power system transition between 1990 and 2014 under various input assumptions. This is also observed from the sensitivity analysis (Table 7), that small deviations of input parameters result in large differences in solutions.

Given Cameroon's strategic location, the nation aims to establish itself as both an electricity exporter to its northern and western neighbors (Nigeria and Chad), and as a transit point between its southern neighbors (Republic of the Congo and Gabon) and the said northern and western neighbors [29,45]. On the former, the nation aims to exploit its abundance hydropower potential. Activities towards both targets have already commenced. Cameroon also plans grid interconnection with other countries in line with creating the Central African Power Pool (CAPP) [33]. However, only national demand and capacity are considered in this study. This is due to export targets or transport capacities not yet quantified, and the obligatory constraint to meet an equilibrium in LEAP/NEMO [36].

Also, the inconsideration of the rebound effect and interactions of the various policies, coupled with the assumption that the heterogeneous actors in the real world would always act rationally, which is hardly the case are other limitations of this study. Thus, the scenarios evaluated in this study are meant to be illustrative, i.e., "for insights, not numbers", so should be interpreted with caution. Moreover, the "the energy system is highly complex and it cannot be steered to a single least cost state anyway" [63]. Suggestions for future study include the use of improved bottom-up energy modelling tools that aim to more-realistically represent the energy transition, better handle uncertainties, or at least diverge from cost optimal approaches under perfect foresight [63]. Others

Table 8
Change in the results of combination scenarios (%).

	Scenario 1	Scenario 2
Capacity in 2045	-46	-45
Cumulative costs	-25	-24
Cumulative GHG emissions	-29	-42

include assessing the flexibility, hosting capacity and grid integration impacts of renewable energy sources, particular solar PV and wind.

6. Conclusion and policy implications

The dire state of Cameroon's power sector and current electricity crisis could primarily be attributed to the generation-demand imbalance. Amongst the many approaches the state has explored, most have focused on increasing generation capacity. However, implementing upstream demand reducing strategies provide a viable and sustainable alternative compared to generation increase.

This study proposed three alternative implementable approaches: Power losses, demand-side energy efficiency and emission targets which could reduce upstream demand and sector emissions. These alternatives were modelled as least-costs pathways based on two levels of implementation. The alternative least-costs pathways were assessed, and in comparison, to the BAU scenario between 2016 and 2045. Results indicate substantial savings in installed capacity, costs and avoided GHG emissions could be achieved depending on the state's commitments.

Depending on the extent of demand-side energy efficiency measures, up to 41% of planned installed capacity and 21% of associated financial costs could be avoided in 2045. However, power losses reduction holds the most benefits for the Cameroon generation system. Compared to the BAU scenario, over 44% of planned installed capacity could be avoided with economic benefits of about 25% if power losses could be reduced to the world average of 8% [23]. Furthermore, significant GHG emissions savings could also be achieved by power losses and implementing energy efficiency measures. Thus, the state should prioritize energy efficiency measures and reduction of the over-the-top network losses.

The results further illustrate that the Cameroon power sector is a low GHG emitter. The generation system will slowly face out thermal generation and uptake more renewable energy technologies if its expansion is based on least costs. Renewable energy would play a significant role in transitioning to a green power sector and its adoption should thus be encouraged in Cameroon. This is also necessary to reduce the dependence on hydropower and ensure higher energy diversity and security. Also, the country should aim at 100% carbon free power system if hopes to significant benefit from carbon taxes and emission savings given its low potentials. This would however entail a turnaround in some of her policies. Such policies include the planned expansion of natural gas generation in the Energy Sector Development Project (PDSEN) and biomass plant in the NDCs [2,7].

Recommended energy efficiency measures for the Cameroon power sector include: appliance and equipment standardization, development of building energy codes for residential and commercial buildings, demand-side management (DSM) programs, mandatory energy audits, and rolling-out education and behavior modification programs for its population, as stated in the NEEP [65–67]. Recommended additional measures not stated in the NEEP include: providing financial incentives and programs, outreach and coalitions with the industry, and decoupling the utility business model [68]. The latter, an electricity tariff adjustment mechanism that severs the linkage between amount of energy usage and utility revenue. This can be done through the introduction of appropriate utility business models to replace the current vertically integrated electricity market. Furthermore, the state should encourage the deployment of roof-stop solar PVs to optimize energy balances of buildings.

Recommended actions for minimizing power losses in Cameroon entails conducting appropriate power system studies to determine/select the optimal size, location and specifications of grid components (such as transformers, transmission and distribution lines, capacitor banks and compensators). These measures however aim at reducing technical losses. Given non-technical losses make up the bulk of the network losses, key actions should be the mandatory national deployment of prepaid meters for the residential and commercial sectors. Advanced monitoring infrastructure (AMI) should also be upscaled.

Summarily, implementing the three alternative pathways, particularly the energy efficiency and power losses scenarios leads to reduced new installed capacity compared to the BAU scenario. This capacity and associated costs savings could be instrumental in ensuring the sustainable development of Cameroon through:

- The decoupling of Cameroon's economic growth and its energy demand growth. This is essential if the nation wants to minimize the amount of resources needed for its industrialization, while meeting its NDC objectives and reducing ecological scarcity [69].
- The reduced pressure on the generation side to provide clean, affordable and modern electricity access. Stakeholders thus gain more time available for rational energy planning and formulation of an energy policy for Cameroon.
- Substantial economic savings from avoided investment, O&M, fuel costs from new power plants.
- Increased energy security for Cameroon from the reduced importation of fossil fuels for new thermal plants.

The future research would consider a higher time and temporal (spatiotemporal) resolution to better represent the interaction between the short-term behavior of renewables and long-term generation expansion planning. The future study should also consider the whole energy system analysis of Cameroon.

Funding

This study was funded by the Regional Center for Energy and Environmental Sustainability (RCEES) of the University of Energy and Natural Resources, Sunyani-Ghana and it is therefore acknowledged.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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